



Hippo training prerequisites

Matthew Falcone

- Apptainer
 - Installation
 - Usage
 - Hippo's release container
 - Commands: shell, exec, run
 - Using MPI: hybrid model vs how we will use it
- Hippo training materials
- Pyvista
 - Unsurprisingly, all postprocessing is done in Pyvista
 - This can be used inside the container or in a separate virtual env

Apptainer: overview

- Apptainer is a container platform created with HPC in mind
- Like other containers, it packages an applications, its libraries, dependencies
 - Reproducible across different systems
- Hippo has a lot of dependencies
 - OpenFOAM: we also need to apply a patch to
 - MOOSE: installation known to be a bit of slog
 - Ask Kingsley for more of a rant
 - We would be spending all morning doing installation if not
- You may have heard of Singularity
 - Apptainer is the community driven successor maintained by the Linux Foundation
 - The split is not too dissimilar to OpenFOAM: .org vs .com

Apptainer: installation

- Hippo's release container is built using Apptainer 1.4.5
 - It will probably work on other versions
- To install on Debian/Ubuntu:
 - Windows users should use WSL

wget https://github.com/apptainer/apptainer/releases/download/v1.4.5/apptainer_1.4.5_amd64.deb

sudo apt install -y ./apptainer_1.4.5_amd64.deb

- If you use MacOS you will need to install a virtual machine such as Lima
 - Ask me if you have any issues

Using Apptainer

- For convenience, we will mostly be using the apptainer shell command
 - OpenFOAM has a lot of commands for pre- and post-processing simulations
 - Mesh generation, partitioning etc.
- Using MPI
 - Apptainer can run MPI within the container or in hybrid mode
 - Hybrid mode is where MPI is launched from the host
 - HPC often system have there own communication backends optimised for their system
 - Hybrid mode can take advantage of this
 - We will use MPI launched from the shell
- Apptainer mounts your home directory into the container
 - We clone the training repo into your home directory

Hippo's release container

- Contains debug and opt installations of Hippo and dependencies
 - Hippo: located at /opt/hippo
 - OpenFOAM-12 (org): located at /opt/openfoam
 - MOOSE: located at /opt/moose - heat_transfer and thermal_hydraulics modules compiled

- Pulling the container

```
apptainer pull hippo.sif oras://quay.io/ukaeea/hippo:v0.1.0-rc2
```

- Running the containers shell

```
apptainer shell hippo.sif
```

- Running other commands

```
apptainer exec hippo.sif <insert shell commands here>
```

- Running Hippo from the host

```
apptainer run hippo.sif -i main.i # equiv to hippo-opt -i main.i
```

Hippo training materials

- Can be downloaded at:

git clone https://github.com/mattfalcone1997/hippo_training_material.git

- Contains material for conjugate heat transfer and thermal-hydraulic coupling
- In breaking news, all postprocessing is done in Pyvista
 - Includes requirements.txt to install it and its dependencies

